

Polydice: Delivering mobile, personalized recipe-sharing

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Using Google Cloud Platform, Polydice achieved greater and efficiency, and costs, while establishing a strong foundation to become an AI-led

ABOUT POLYDICE



Polydice

About Polydice

Founded in 2010, Polydice operates the iCook social recipe website and application. iCook is one of the most popular recipe services in Taiwan with approximately 5 million visitors per month and approximately 4.5 million downloads.

Google Cloud Results

- Supports 5 million users per

Deliver
improved
data

Industries: Gaming, Media & Entertainment

Location: Taiwan

month and
4.5 million
downloads

- Enables rapid response to social media user queries
- Provides a foundation to become an AI-led company

infrastructure
stability

BigQuery (<https://cloud.google.com/bigquery/>),

Cloud SQL (<https://cloud.google.com/sql/>),

Cloud Storage (<https://cloud.google.com/storage/>)

(<https://cloud.google.com/memorystore/>), AI

Cloud Composer

(<https://cloud.google.com/composer/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Memorystore

(<https://cloud.google.com/memorystore/>)

Firebase (<https://firebase.google.com/>),

AI Platform (<https://cloud.google.com/ai-platform/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

Dialogflow Enterprise Edition

(<https://cloud.google.com/dialogflow-enterprise/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

Firebase (<https://firebase.google.com/>)

Google Workspace

(<https://workspace.google.com/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Calendar

(<https://workspace.google.com/products/calendar/>)

Cloud Identity (<https://cloud.google.com/identity/>)

In Taiwan smartphones are increasingly popular as people look to manage their personal and business activities. As part of the trend, tech entrepreneurs Richard Lee and Fong Hsiao, who younger people were often taking photos of their kitchens to interact with as they prepared meals. Lee and Hsiao decided to create a social network to enable people to use their smartphones to find and share recipes.

The team, who established the local food delivery business Polydice (<https://polydice.com/>), launched iCook (<https://icook.tw/>) in 2010. Today, iCook is one of the most popular recipe services in Taiwan, with 4.5 million visitors per month and 4.5 million downloads. Lee and Hsiao now aim to extend iCook to other markets and its supplementary online products, including a kitchenware and ingredients ecommerce site and a

site featuring blogs prepared by in-house editors – beyond Taiwan to residents of Hong Kong and other countries.

Lee points to the technical skill and experience within Polydice as integral to iCook's growth and success. About half of the business's 50 workers are engineers, product managers, and designers.

Polydice has also used several Google solutions and services to develop and grow its products. These include using Firestore (<https://firebase.google.com/>) to undertake mobile application development; making iCook available through Google Play

(<https://play.google.com/store>); posting cooking videos on YouTube; and reaching consumers through Google Ads (<https://ads.google.com/home/>).

Polydice was also already an extensive user of Google Workspace (<https://workspace.google.com/>), including Gmail

(<https://workspace.google.com/products/gmail/>) for interviews, Calendar

(<https://workspace.google.com/products/calendar/>) to manage appointments and events, and Google Meet

(<https://workspace.google.com/products/meet/>) for voice and video meetings. "We use several Chrome devices throughout the company, including Chrome Sticks we plug into televisions to review dashboards in meeting rooms," Lee says. The business also trials mobile device management and user authentication through Cloud Identity

(<https://cloud.google.com/identity/>).

“Services such as Cloud AutoML, which enable developers with minimal machine learning experience to train high-quality models, mean businesses like ours don’t need to hire data scientists proficient in machine learning algorithms. We can train product managers in machine learning and give them access to Cloud AutoML to

train models
themselves.”

—Richard Lee, Co-founder
and Chief Technology
Officer, Polydice

Polydice started running iCook in a traditional cloud. However, as the website and application became more popular, the business decided to evaluate cloud services that could better meet its long-term infrastructure, data analytics, and application needs. “We started experiencing problems, such as data queries becoming stuck, delaying resolution and completion of other queries,” explains Lee. “We would have to assign engineers once or twice a month to resolve these issues. This threatened to slow the timeliness and effectiveness of decision-making.”

Superior data services

Polydice’s experience with Google made reviewing [Google Cloud Platform](https://cloud.google.com/) (https://cloud.google.com/) an obvious choice. “We found Google Cloud Platform provided superior data services at a lower cost than the data warehouses we had been using,” says Lee. “Our analysis also revealed Google infrastructure and application services were better than products offered by its competitors.”

Google’s decision to locate a Google Cloud Platform Region in Taiwan also played a key role in the

business' evaluation. "The Taiwan location meant we could reduce iCook application response times to local users while keeping sensitive customer data on-shore to comply with regulatory requirements," explains Lee.

AI and ML provide new horizons

Polydice's founders believed Google's visionary approach to machine learning and artificial intelligence could open up new horizons for the business. "We saw we could benefit considerably from Google's approach," says Lee. "For example, viewing tutorials and case studies on Google's websites gave us ideas about how artificial intelligence (AI) and machine learning (ML) could help Polydice and iCook grow."

Polydice then seamlessly migrated iCook from its incumbent cloud service to Google Cloud Platform.

The business now uses BigQuery.

(<https://cloud.google.com/bigquery/>) to provide serverless, scalable data warehousing and Cloud Storage (<https://cloud.google.com/storage/>) to store objects. Cloud Composer

(<https://cloud.google.com/composer/>) provides the workflow orchestration services Polydice uses to run pipelines across Google Cloud Platform and other cloud environments.

“Overall, Google Cloud Platform provides superior stability, efficiency, and cost savings compared to our previous services. I would describe the caliber of service, particularly data infrastructure stability, as 10 times better.”

—Richard Lee, Co-founder
and Chief Technology
Officer, Polydice

The business employs Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) cluster management and orchestration services for its containerized applications, including iCook and its supplementary properties, and Cloud SQL (<https://cloud.google.com/sql/>) to run its relational

database. [Cloud Memorystore](#)

(<https://cloud.google.com/memorystore/>) for Redis delivers a managed in-memory data storage service that Polydice uses to create application caches.

Polydice also uses a [Virtual Private Cloud](#)

(<https://cloud.google.com/vpc/>) to provide managed networking for its Google Cloud Platform resources and to protect back-end services from public access. “We are working on migrating our Kubernetes Engine cluster into a Virtual Private Cloud as we believe the private cloud service is more secure and provides greater flexibility to configure networks,” says Lee. “We also like the Alias IP feature that makes it easier for us to run multiple services on virtual machines.”

Delivering social media success

Polydice uses [Cloud Machine Learning Engine](#)

(<https://cloud.google.com/ml-engine/>) to help deliver trend and prediction services, and [Cloud Functions](#)

(<https://cloud.google.com/functions/>) to develop and scale code in the cloud. “We used Cloud Functions to build a machine learning model that enables us to manage our social media presence,” explains Lee.

“For example, we have a Facebook page liked by more than two million people. A lot of our fans come to this page to ask questions about our recipes, such as ‘where can I buy the ingredients?’ or ‘what ingredients can I combine to create a delicious meal?’”

“We want to
become an AI-
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Google gives us
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to achieve that
goal.”

—Richard Lee, Co-founder
and Chief Technology
Officer, Polydice

Polydice subsequently moved the model to Cloud AutoML (<https://cloud.google.com/automl/>) to take advantage of improved performance and automation. “Services such as Cloud AutoML, which enable developers with minimal machine learning experience to train high-quality models, mean businesses like ours don’t need to hire data scientists proficient in machine learning algorithms,” says Lee. “We can train product managers in machine learning and give them access to Cloud AutoML to train models themselves.”

Cloud AutoML has elevated the creativity levels of an already innovative business. “We came up with several ideas that our salespeople believed could help with predictions and lead generation if the right data was fed through Cloud AutoML,” Lee says.

The business is also using Cloud Dialogflow (<https://cloud.google.com/dialogflow-enterprise/>) and Cloud Functions to build a personal recommendation engine accessed through a chatbot interface. “If we can track people’s preferences, we can provide more personalized, relevant services to them,” says Lee. “For example, if we know they’re keen on fried snacks at midnight, we can guide them to the right recipes to satisfy their hunger!”

Query bottlenecks eliminated

Google Cloud Platform eliminated the query bottlenecks that started to plague the business in previous environments. “We’ve not experienced any stuck queries that delayed other queries since moving to BigQuery,” explains Lee.

“Overall, Google Cloud Platform provides superior stability, efficiency, and cost savings compared to our previous services. I would describe the caliber of service, particularly data infrastructure stability, as 10 times better.”

An AI-driven company

With its Google Cloud Platform infrastructure, applications, and data analytics environments well established, Polydice is now looking to the future. The business plans to develop a voice assistant based on Cloud Dialogflow for iCook and to create a ML-based recipe image classification system.

“Google Cloud gives us many tools to tackle problems and take advantage of new opportunities,” Lee says. “We might have previously said we wanted to become a data-driven company, but now we want to become an AI-driven company. Google gives us the foundation to achieve that goal.”